CLAIMS

What is claimed is:

1. A mass storage system, comprising:
   - a high-radix storage medium comprising a plurality of storage elements, wherein each storage element is configured to reliably hold one of N discrete, distinguishable physical states, where N is an integer greater than two; and
   - a high-radix storage controller communicatively coupled to the high-radix storage medium, the controller configured to:
     - receive abstract symbols for writing;
     - translate said abstract symbols into corresponding physical states from the N discrete states; and
     - cause the high-radix storage medium to store said physical states in its storage elements.

2. The mass storage system of claim 1, wherein the high-radix storage controller is further configured to:
   - read physical states from the storage elements of the high-radix storage medium; and
   - translate said physical states back into abstract symbols.

3. The mass storage system of claim 1, wherein the storage elements are flash memory cells configured to store N distinct charge levels.

4. The mass storage system of claim 1, wherein the storage elements are magnetic domains configured to store N distinct magnetic orientations.

5. The mass storage system of claim 1, wherein the abstract symbols represent unevaluated mathematical expressions.

6. A method for storing data in a mass storage system, comprising:

- providing a high-radix storage medium comprising storage elements, each element capable of reliably holding one of N discrete, distinguishable physical states, where N is an integer greater than two;
- receiving abstract symbols at a high-radix storage controller;
- translating said abstract symbols into corresponding physical states from the N discrete states;
- causing the storage elements of the high-radix storage medium to assume said physical states, thereby storing the data; and
- subsequently, reading the physical states from the storage elements and translating them back into the abstract symbols.

7. The method of claim 6, wherein the received abstract symbols are components of an unevaluated parametric data structure.

8. The method of claim 6, wherein N is 16 or greater.